EMILY JIA

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STATEMENT

I am Emily Yue-ting Jia, a graduate student at the University of Southern California, advised by **Prof. Yajie Zhao** and **Prof. Yue Wang**. I received my bachelor's degree from Tsinghua University. My current research focuses on physics-grounded video generation and 3D generation, with an emphasis on incorporating physical laws into generative models and learning physical dynamics from data. Previously, I conducted research on 3D scene reconstruction under the guidance of **Prof. Yushen Liu**.

Besides research, I also spend a lot of time on software engineering. I was a leader in the Network department of Tsinghua CS student technology group. I have plenty of experience in website development and tutoring in website development, as part of our group activity.

EDUCATION

Ph.D in Computer Science, University of Southern California — August 2023 - Now
GPA: 3.94

Undergraduate in Computer Science and Technology, Tsinghua University — August 2019 - Jul 2023
GPA: 3.79

SKILLS

Research	Computer Vision, Physics-grounded Learning, Video Generation, 3D Generation
Engineering	Android, Front-end, Database
Tools	PyTorch, PyFlex, SoftGym, Blender
Language	Python, C++/C, JavaScript

SCHOLARSHIP

I'm awarded with Viterbi School of Engineering Fellowship for Incoming Students in USC. And in Tsinghua, I'm awarded as Excellent International Student by Beijing government from 2019 to 2022.

RESEARCH PROJECTS

Project 1 Learning an Implicit Physical Model for Image-based Fluid Simulation

- with **Prof. Yue Wang** and **Prof. Yajie Zhao**
- **ICCV 2025**
- **Project Page: https://physfluid.github.io/**
- We present a new method capable of recovering the 4D scene from a single image and generating physically plausible videos that capture both fluid motion and camera movement.
- We propose physics-informed neural dynamics, a novel model that can predict physics-grounded flow dynamics from a single image of natural fluids.
- Experiments show that our framework surpasses baseline methods, producing more realistic animations, particularly in image editing tasks where the boundary geometry is changed.

Project 2 Seeing the Wind from a Fallen Leaf

- with **Prof. Yue Wang**
- **Neurips 2025**
- **Project Page: https://chaoren2357.github.io/seeingthewind/**
- We propose a novel self-supervised framework for learning invisible dynamics from videos, bridging the gap between human vision and computer vision.

- We introduce a suite of techniques that enable robust and stable learning of forces across objects with diverse materials, including real-world captured objects, achieving effective force estimation
- We demonstrate potential applications, such as physics-based video synthesis, showcasing the practical value of our approach.

Project 3 DreamPlan: Efficient Reinforcement Fine-Tuning of Vision-Language Planners via Video World Models

- with **Prof. Yue Wang**. Submitted to **IROS 2026**
- VLMs possess semantic priors and reasoning ability but lack physics knowledge, which is crucial for robotic tasks, especially deformable manipulation.
- To inject task-specific physics knowledge to VLMs, RL fine-tuning is needed. However, RL requires massive interaction data, which is expensive to collect on real-world robots. To overcome this bottleneck, we introduce DreamPlan, a novel world-model-assisted RL framework designed to efficiently align VLM planners with real-world physics via model-generated virtual rollouts.

Project 4 NeRFPrior: Learning Neural Radiance Field as a Prior for Indoor Scene Reconstruction

- from summer 2022 to spring 2023, with **Prof. Yushen Liu**
- **CVPR 2025 Highlight**
- **Project Page: https://wen-yuan-zhang.github.io/NeRFPrior/**
- We propose a 2-stage training pipeline for faster and more detailed 3D scene reconstruction. In the first stage, we use a voxelized neural network to approximate the radiance field. In the second stage, we input the coarse radiance field as prior into a SDF-based Nerf and get a consistent and fine-grained field for mesh reconstruction.
- Experiments show that our method achieves better reconstruction performance with less training time, compared with previous method such as Neus and unisurf. Meshes retrieved from our method have flatter wall and floor areas and more accurate shapes for small crafts such as bowls and spoons on the table.

Other Research Projects

- PhysAgent++: Grounding AI Agents into the Real World with Physics-Aware Reinforcement Learning
  - with **Prof. Yue Wang**. Submitted to CVPR 2026.
  - Proposed a world-model–based reinforcement learning framework that enables LLM-driven agents to learn physics-aware decision-making and long-horizon planning, achieving strong performance on real-world robotic assembly tasks.
- A Polarized Reflection and Material Dataset of Real World Objects
  - with **Prof. Yajie Zhao**.
  - **CVPR 2026**
- DreamSurfels: Single-Image 3D Generation with Geometry-Conditioned Video Diffusion
  - with **Prof. Yajie Zhao**. Submitted to NeurIPS.
  - Proposed a 3D object generation methdo with explicit geometry conditions and joint diffusion model.

OTHER PROJECTS

Project 1 Chrome Recorder Extension

- A chrome extension enables the recording of clicking, copying, inputting, and page capture.
- GitHub: Chrome Recorder Extension
- used by Tsinghua NLP lab for data collection.